IGVC - Intelligent Ground Vehicle

UTA Department of Computer Science and Engineering

Senior Design

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Background

IGVC is an annual competition held to challenge university students to design and implement a fully autonomous ground vehicle. Our robot utilizes a Jetson Orin NX to perform real time decision making using camera vision and LiDAR detection. Using this, our vehicle can navigate through a series of obstacles in an outdoor environment. The system addresses real world applications in autonomous mobility and intelligent ground vehicles.

Key Requirements

Human driving mistakes, labor shortages, and unsafe environments are major reasons why autonomous navigation is increasingly needed in real-world applications. Integrating autonomy in these fields will increase safety and efficiency all around. Modern processors can execute millions of instructions a second, leading to rapid decision making based on precise sensor readings. Our vehicle uses this capability to process real-time information and calculate distance, speed, and obstacle measurements to guide us through the unknown.

Architectural Design

Input Layer

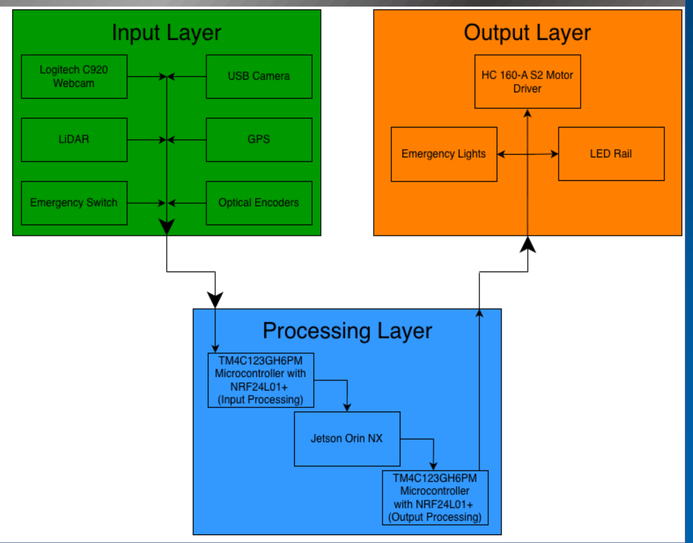
- LiDAR and Cameras collect real time information to set vehicle's overall direction and speed
- Optical Encoders and GPS refine the vehicle's position and movement for accurate navigation
- Emergency Stop Switch is used to cut power off to the motor in case of unexpected situations

Processing Layer

- TM4C (input) takes in information from GPS and Encoders and passes it to the Jetson Orin
- The Jetson Orin NX processes data from LiDAR, camera vision, and TM4C to determine the direction/speed of the motors and emergency light sequence.
- TM4C (output controller) receives these commands and sends exact signals to the motor drivers.

Output Layer

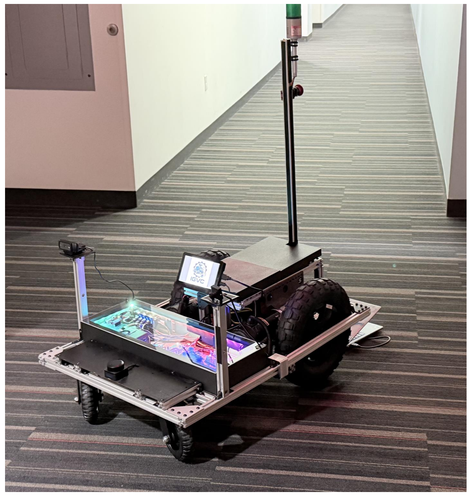
- HC 160-A S2 motor drivers deliver the correct power and direction to the motors, moving the vehicle
- Emergency Lights are activated based on the state the machine is in, alerting anybody nearby.

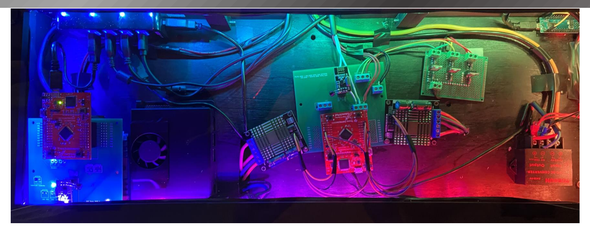


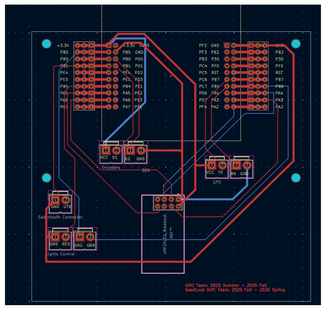
Implementation Details and Test Plan

Implementation details and test plan

- Lane detection pipeline tested with real-time two camera inputs.
- LiDAR obstacle detection validated during field tests.
- HC 160-A SC motor drivers for driving the motors with a 26V battery.
- Motor Controls with TM4C123GXL board. NRF24L01+ facilitating the communication between the Jetson Orin NX and Motor Drivers.
- Emergency-stop tested with wireless NRF24L01+ module and STM32F446RE.
- Indicator lights and Brake lights connected with Custom built PCB's using MOSFETs.







Conclusions and Future Work

The IGVC annual competition will take place in the summer of 2026. Our senior design team, working from Fall through Spring, will continue to develop the project, building upon the existing work and adding new functionalities to ensure we meet the competition's requirements. One exciting addition we plan to integrate is hydraulic systems, with the goal of enabling the vehicle to perform a hopping function.

References

1. S. McDermott, “Motor maniacs: Autonomous ground vehicle senior design project,”https://cse.uta.edu/senior-design/projects/motor-maniacs/, 2025, university of Texas at Arlington, posted May 5, 2025. Accessed: 2025-09-19.
2. Intelligent Ground Vehicle Competition 2026 Rulebook, 2026, https://www.igvc.org.
3. Texas Instruments, “Tiva™ C Series TM4C123GH6PM Microcontroller Data Sheet,” https://www.ti.com/lit/ds/symlink/tm4c123gh6pm.pdf, accessed 2025-10-10.

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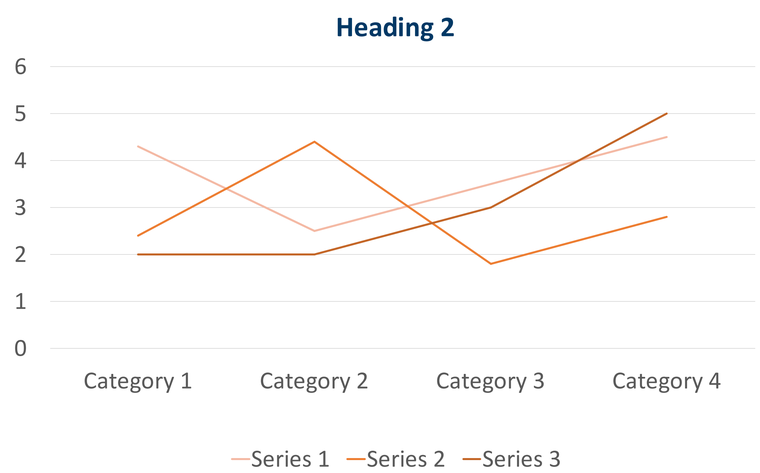
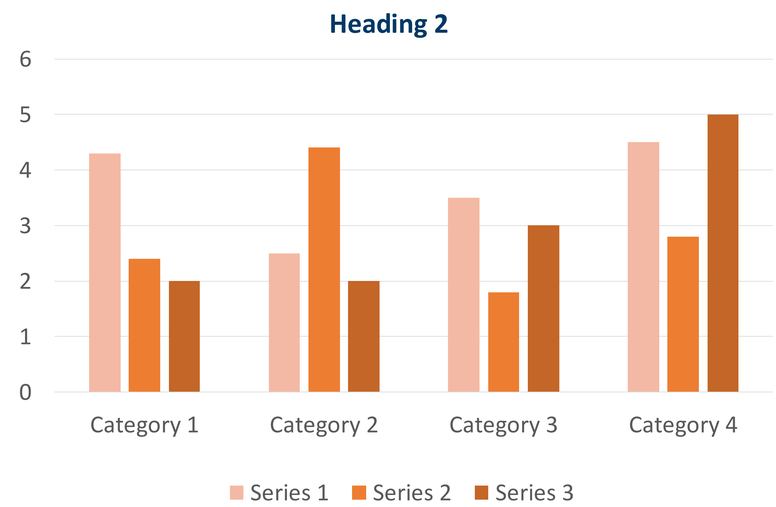
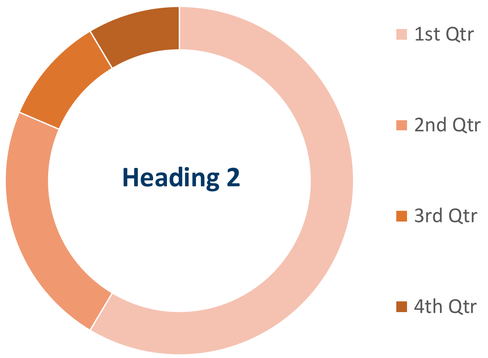
Charts

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